

Protocol for open-ended registration following the Preferred Reporting Items for Systematic Reviews and Meta-Analyses Extension for Scoping Reviews (PRISMA-ScR) guidelines provided by Tricco and colleagues (2018)

Title: A scoping review of neuroimaging evidence concerning semantic representations in multilingual speakers

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Structured summary: The neural underpinnings of semantic representations involve a distributed network of cortical regions that integrate multimodal information relating to concepts. These semantic representations are formed dynamically through novel experience and information, including linguistic. Most models of semantic knowledge and its structure in the brain have been based on WEIRD (Heinrich, Heine, & Norenzayan, 2010), monolingual populations that fail to capture the rich and diverse multilingual experience that is the reality for a majority of the global population. While the influence of multilingualism on cognition has been studied for decades, the bulk of studies have used behavioral methods across a wide range of tasks, and there is a dearth of theory-driven and reproducible research. Our scoping review seeks to summarize, quantify, and critically appraise existing functional neuroimaging-based research on semantic representations in healthy adult multilingual speakers. Five databases will be searched and carefully charted to understand the extent of research in this domain and to identify convergence across studies regarding differences in the neural code of semantic representations in multilingual speakers. Through this effort, we will be able to clearly understand limitations in current research approaches, identify gaps in the literature, and offer insights and recommendations for future investigations in this area.

Rationale: When we move about the world, we perceive and interact with a wide range of objects in our physical and social environment. The resultant semantic representations are internal and adaptable models that allow us to predict outcomes and update our knowledge when presented with new information. Through language, we label concepts, which allows for direct associations between other semantic representations. The semantic network tying these various representations together is constantly evolving with novel experiences and knowledge, including linguistic (e.g., additionally acquired languages). Several models of conceptual knowledge have been put forward, emphasizing the contribution of category (Collins & Loftus, 1975; Damasio, 1989) or featural overlap (Binder et al., 2016; Martin, 2016).

The largest convergence of evidence across experimental, neuropsychological, and computational studies supports a partially grounded (Barsalou, 2008; Binder & Desai, 2011; Kiefer & Pulvermüller, 2012) model that engages a distributed network of modality-specific and supramodal regions spanning the cortex (Patterson & Ralph, 2016). However, and as is the case with many theories in cognitive neuroscience, these models have been tested primarily on WEIRD (Heinrich, Heine & Norenzayan, 2010), monolingual populations that do not properly represent the reality of our global population.

Examination of multilingual cohorts in experimental psychology (Ransdell & Fischler, 1987) and cognitive neuroscience (Klein et al., 1995) has been ongoing for decades. Many researchers have investigated whether multilingual speakers have a performance advantage in cognitive tasks compared to monolinguals (Paap, Johnson, & Sawi, 2015; Grundy, 2020). The lack of consistency across samples, methodologies, and theory-driven experimentation has led to the current, rather opaque state of evidence regarding the impact of multilingualism on cognition (Blanco-Elorrieta & Caramazza, 2021; de Oliveira & Oliveira, 2022). Moreover, more researchers have investigated semantic memory and structure in bilingualism through theoretical or behavioral approaches (Weinrich, 1953; Paivio & Desrochers, 1980; de Groot, 1992; Kroll & Stewart, 1994) than through neuroscientific means, including the application of functional magnetic resonance imaging (fMRI), electroencephalography (EEG), and magnetoencephalography (MEG) recording (Correia et al., 2014, 2015; Van de Putte, 2018).

The rationale for this review stems from both the lack of consistency in theory and approach within the field of bilingual research (de Oliveira & Oliveira, 2022) and the short timespan in which advanced neuroscientific techniques such as representational similarity analysis (Kriegeskorte et al., 2008) have been available for neuroscientific investigations into multilingual semantic representation. It is thus imperative to synthesize and critically appraise all existing research that has used functional neuroimaging techniques to study multilingual semantic representation in order to move the field towards a clear and robust theory of semantic structure in multilingual contexts.

Objectives: The present scoping review aims to consolidate existing neuroimaging studies investigating semantic knowledge in multilingual speakers to provide an overview of the amount and extent of existing research in this area. We will include any study that has used electroencephalography, magnetoencephalography, or functional magnetic resonance imaging in the investigation of semantic representations in multilingual cohorts. Subsequently, we plan to offer insight and considerations for future research examining the neural structure of semantic representations in multilingual speakers.

The following research questions will be addressed in our review:

1. How many studies have utilized functional neuroimaging techniques to explore the contribution of multilingualism to semantic representations?
2. Which paradigms and imaging techniques have been applied in the examination of semantic representations in multilingual speakers?
3. How has the multilingual experience been quantified across these studies?
4. What functional neuroimaging results have been associated with multilingualism in these studies, and to what extent do these findings converge?

Protocol & Registration: The current protocol was drafted according to the Preferred Reporting Items for Systematic Reviews and Meta-Analyses Extension for Scoping Reviews (PRISMA-ScR) Checklist (Tricco et al., 2018) and revised by the research team. This protocol was registered to the corresponding project within the Open Science Framework (OSF) on February 15, 2023. Registration occurred prior to starting the review process.

Eligibility Criteria:

Inclusion Criteria

- Experimental studies applying fMRI, MEG, or EEG
- Experimental tasks probing semantic knowledge
- Studies investigating healthy multilingual adults without a history of neurological or psychiatric disorders

Exclusion Criteria

- Studies applying any other technique apart from fMRI, MEG, or EEG
- Experimental tasks that do not probe semantic knowledge
- Studies investigating monolingual, clinical, or pediatric populations

Information Sources: We will utilize the following electronic databases to identify relevant literature for the scoping review:

- Web of Science Core Collection (<https://www.webofscience.com/>)
- PubMed (<https://www.ncbi.nlm.nih.gov/>)
- PsycINFO (<https://psycnet.apa.org/>)

Search Strategy: The collection of relevant literature will start in February 2023 and will include all articles published before February 20, 2023. Given the narrow scope of this research topic and the emergence of neuroimaging and neurophysiological techniques only in the past few decades, all relevant articles will be included, regardless of publication year. Gray literature in the form of conference proceedings, scientific reports, doctoral dissertations, and pre-registered pre-prints will be included in the search. The search strategy will be documented in accordance with the PRISMA-S: PRISMA Search Reporting Extension (Rethlefsen et al., 2021). An example keyword search within PubMed is presented in Table 1 for the following keywords: ("fMRI" OR "functional magnetic resonance imaging") OR ("MEG" OR "MAGNETOENCEPHALOGRAPHY") OR ("EEG" OR "ELECTROENCEPHALOGRAPHY") AND ("BILINGUA*" OR "MULTILINGUA*") AND ("SEMANTIC*" OR "CONCEPT*").

Table 1. Search example in PubMed: <https://www.ncbi.nlm.nih.gov/>.

Database	Keyword Search	Filter(s)	Year Range
PubMed	("fMRI" OR "functional magnetic resonance imaging") OR ("MEG" OR "MAGNETOENCEPHALOGRAPHY") OR ("EEG" OR "ELECTROENCEPHALOGRAPHY") AND ("BILINGUA*" OR "MULTILINGUA*") AND ("SEMANTIC*" OR "CONCEPT*")	Adult: 19+ years	1990-2023

Selection of Sources of Evidence: Identified citations will be imported into a shared Rayyan (<https://www.rayyan.ai/>) repository and duplicates will be removed. All identified articles will be split into three unique groups, and the three reviewers will independently screen each article title and abstract in two of the three groups. Reviewers will be blinded to the other reviewers' decisions. During evaluation, a checklist pertaining to the eligibility criteria will be used and excluded articles will be marked in the Rayyan repository along with the reason(s) for exclusion. Discrepancies regarding selected articles will be resolved by the third reviewer who has not already screened the article. Following this preliminary selection, full publications, including any supplementary material, will be reviewed by the first author (A.E.L). Information regarding the excluded articles will be explicitly reported in the methods section of the review via the PRISMA flow chart template (Page et al., 2021).

Data charting process: Data will be charted according to Chapter 11 of the Joanna Briggs Institute (JBI) Manual for Evidence Synthesis (Peters et al., 2020). The following information will be extracted for each article: Author(s); year of publication; origin of publication (country where study was published and/or conducted); aims/purpose; study population; study sample size; methodology (experimental paradigm, data collection and analysis); key findings that directly relate to the review questions of the current scoping review. Data charts will be created by author A.E.L. and verified by author V.B. Final data charts, including any revisions by author V.B., will be made available as supplementary information in the manuscript.

Data items: The following variables will be abstracted per article:

- **Publication information:** author(s) name(s), publication year, journal, location of study
- **Cohort demographic data:** age, sex, education, location, size, type and proficiency of languages spoken, age of language acquisition, frequency of language use
- **Paradigm information:** task name, stimulus modality, testing language
- **Methodological variables:** type of neuroimaging (fMRI) or neurophysiological recording (EEG, MEG) and details pertaining to protocol, analytical approach (e.g., whole-brain, or ROI-based and details), information on any open-source code
- **Outcome variables:** imaging- or electrophysiological-related results and author(s) interpretation

Critical appraisal of individual sources of evidence: An appraisal of the charted data will be necessary to provide an overview of convergent and divergent evidence regarding the influence of multilingualism on semantic representations. As such, an in-depth assessment of each article charted will allow the first author (A.E.L.) to group studies according to methodological approach and identify convergence between findings and potential gaps in evidence.

Synthesis of results: Chared studies, with all relevant variables extracted, will be grouped according to methodological approach (e.g., fMRI) and paradigm (e.g., semantic similarity judgment task), while carefully considering differences in cohort demographics and language experience (in the case of cohorts with different language types and/or proficiencies). In synthesizing results, the convergence between findings with regard to brain regions (fMRI) and time windows, amplitudes, or peak latencies (EEG, MEG) will be described in a narrative format. For any studies that explicitly report the coordinate location of findings or quantitative metrics, the results will be synthesized in a table within the scoping review.

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